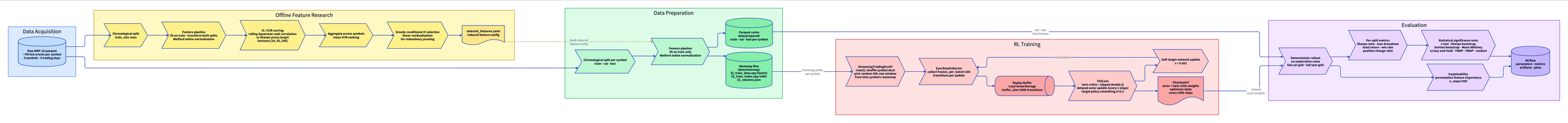

Data Acquisition
Raw MBP-10 parquet
~7M tick events per symbol
6 symbols · 4 trading days
Offline Feature Research
Chronological split
train_size rows
Feature pipeline
fit on train · transform both splits
Welford online normalisation
IC / ICIR scoring
rolling Spearman rank correlation
vs Sharpe-proxy target
horizons [10, 50, 200]
Aggregate across symbols
mean ICIR ranking
Greedy conditional IC selection
linear residualisation
for redundancy pruning
selected_features.yaml
reduced feature config
feeds reduced
feature config
Data Preparation
Chronological split per symbol
train · val · test
Feature pipeline
fit on train only
Welford online normalisation
Parquet cache
data/prepared/
train · val · test per symbol
Memmap files
data/memmap/
{i}_train_data.npy float32
{i}_train_index.npy int64
{i}_columns.json
val + test
DataFrames
memmap paths
per symbol
RL Training
StreamingTradingEnvXY
reset(): shuffle symbol deck
pick random 50k-row window
from that symbol's memmap
SyncDataCollector
collect frames_per_batch=200
transitions per update
next batch
Replay Buffer
LazyTensorStorage
buffer_size=100k transitions
TD3Loss
twin critics · clipped double Q
delayed actor update (every 2 steps)
target policy smoothing σ=0.2
Soft target network update
τ = 0.001
Checkpoint
actor + twin critic weights
optimizer state
every 100k steps
trained
actor weights
Evaluation
Deterministic rollout
no exploration noise
full val split · full test split
Per-split metrics
Sharpe ratio · max drawdown
total return · win rate
position change ratio
Statistical significance tests
t-test · Sharpe bootstrap
Sortino bootstrap · Mann-Whitney
vs buy-and-hold · TWAP · VWAP · random
Explainability
permutation feature importance
n_steps=500
MLflow
parameters · metrics
artifacts · plots